

L'Hopital's Rule

Section 8.7

2-9-26

A) $\frac{0}{0}$

$$\text{Ex: } \lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3} =$$

$$= \boxed{\lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{(x-3)} = 5}$$

Ex w/ LH:

$$\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3} =$$

$$\frac{0}{0} \text{ LH} = \boxed{\lim_{x \rightarrow 3} \frac{2x - 1}{1} = 5}$$

$$\text{Ex: } \lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

$$\frac{0}{0} \text{ LH} = \boxed{\lim_{x \rightarrow 0} \frac{\cos x}{1} = 1}$$

$$\text{Ex: } \lim_{x \rightarrow 2} \frac{\int_2^x \sqrt{t^2 - 1} \, dt}{x - 2} =$$

$$\frac{0}{0} \text{ LH} = \boxed{\lim_{x \rightarrow 2} \frac{\sqrt{x^2 - 1}}{1} = \sqrt{5}}$$

B) $\frac{\infty}{\infty}$

$$\text{Ex: } \lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 5}{4x^2 + 5x - 2} = \boxed{\frac{3}{4}}$$

$$\frac{\infty}{\infty} \text{ LH} = \lim_{x \rightarrow \infty} \frac{6x - 2}{8x + 5}$$

$$\frac{\infty}{\infty} \text{ LH} = \boxed{\lim_{x \rightarrow \infty} \frac{6}{8} = \frac{3}{4}}$$

$$\text{Ex: } \lim_{x \rightarrow \infty} \frac{\ln x}{x^2} =$$

$$\frac{\infty}{\infty} \text{ LH} = \boxed{\lim_{x \rightarrow \infty} \frac{1}{2x^2} = \frac{1}{\infty} = 0}$$

C) $0 \cdot \infty$

$$\text{Ex: } \lim_{x \rightarrow \infty} x \tan\left(\frac{1}{x}\right) =$$

$$0 \cdot \infty \text{ LH} = \lim_{x \rightarrow \infty} \frac{\tan(x^{-1})}{x^{-1}} = \frac{0}{0}$$

$$\frac{0}{0} \text{ LH} = \lim_{x \rightarrow \infty} \frac{x^{-2} \cdot \sec^2(x^{-1})}{-x^{-2}} = \boxed{1}$$

$$\text{Ex: } \lim_{x \rightarrow 0^+} x \cot x =$$

$$0 \cdot \infty \quad \lim_{x \rightarrow 0^+} \frac{x}{\tan x} = \frac{0}{0}$$

$$\frac{0}{0} \text{ LH} \quad \lim_{x \rightarrow 0^+} \frac{1}{\sec^2 x} = \boxed{1}$$

D) $\infty - \infty$

$$\text{Ex: } \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2}\right) = \lim_{x \rightarrow 0^+} \left(\frac{x-1}{x^2}\right) = \frac{-1}{\infty}$$

$$\infty - \infty = -\infty$$

$$\text{Ex: } \lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} - \frac{1}{x}\right) = \lim_{x \rightarrow 0^+} \left(\frac{x - \sin x}{x \sin x}\right) = \frac{0}{0}$$

$$\frac{0}{0} \text{ LH} = \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos x}{\sin x + x \cos x}\right) = \frac{0}{0}$$

$$\frac{0}{0} \text{ LH} = \lim_{x \rightarrow 0^+} \left(\frac{\sin x}{\cos x + \cos x - x \sin x}\right) = \frac{0}{2}$$

$$= \boxed{0}$$

E) $1^\infty, 0^0, \infty^0$

$$\text{Ex: } \lim_{x \rightarrow \infty} x^{1/x} = \infty^0$$

$$\ln y = \lim_{x \rightarrow \infty} \ln x^{1/x}$$

$$\ln y = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \frac{\infty}{\infty}$$

$$\frac{\infty}{\infty} \text{ LH} \quad \ln y = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\ln y = 0$$

$$e^0 = y$$

$$\boxed{y = 1}$$

$$\text{Ex: } \lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x =$$

$$1^\infty \quad \ln y = \lim_{x \rightarrow \infty} \ln \left(1 + \frac{3}{x}\right)^x$$

$$\ln y = \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{3}{x}\right) = \infty \cdot 0$$

$$\ln y = \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{3}{x}\right)}{x^{-1}}$$

$$\frac{0}{0} \text{ LH} \quad \ln y = \lim_{x \rightarrow \infty} \frac{\frac{-3}{1+3x^{-1}}}{-x^{-2}}$$

$$\ln y = \lim_{x \rightarrow \infty} \frac{3}{1+3x^{-1}}$$

$$\ln y = 3 \quad \boxed{y = e^3}$$

Patterns:

$$\lim_{x \rightarrow \infty} \left(1 + \frac{5}{x}\right) = e^5$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^{2x} = e^{3 \cdot 2}$$